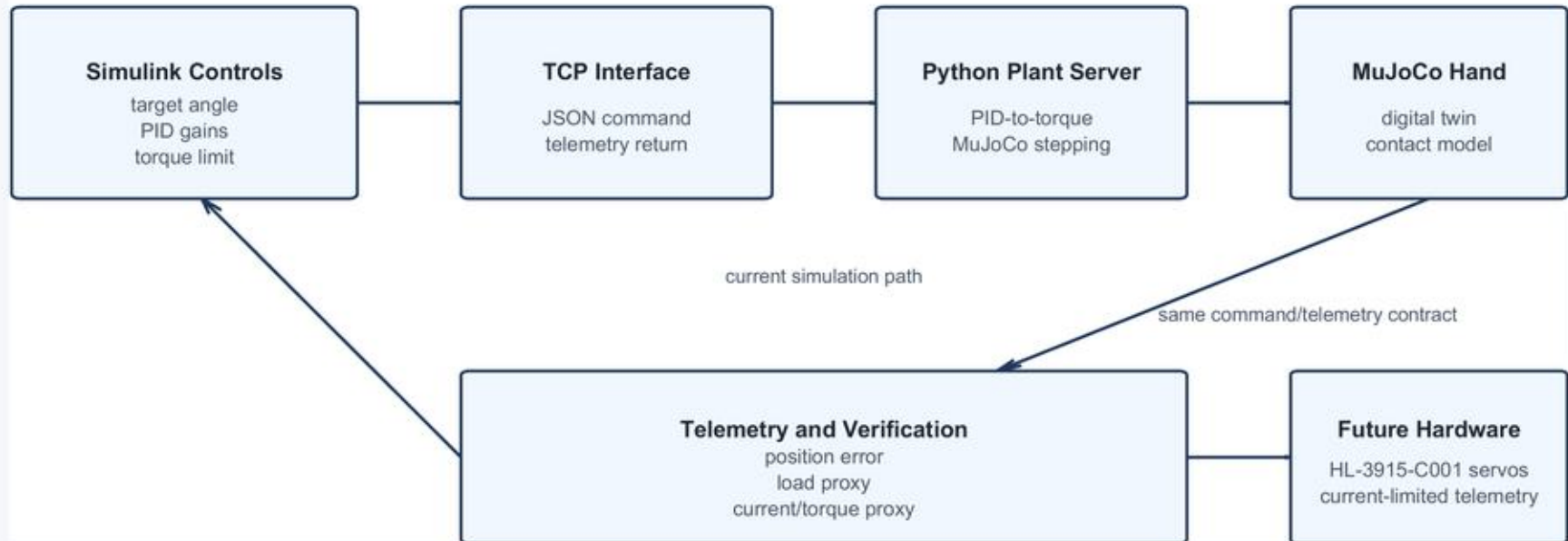


Simulation-to-Real Control Workflow

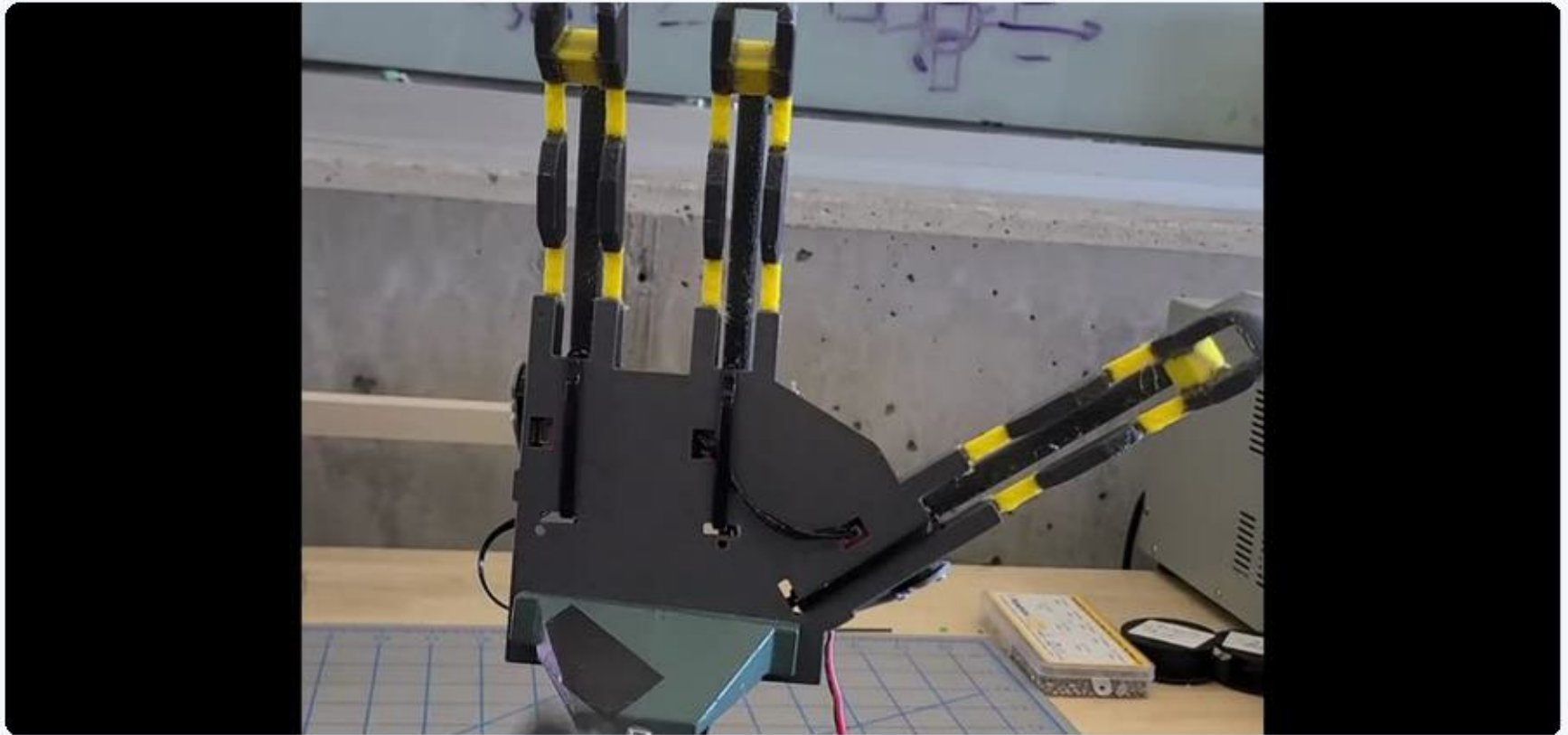
Simulink tunes the controller; MuJoCo acts as the plant; the interface is shaped for hardware transfer.



PORTFOLIO OVERVIEW

Robotics R&D from mechanism concept to validated hardware

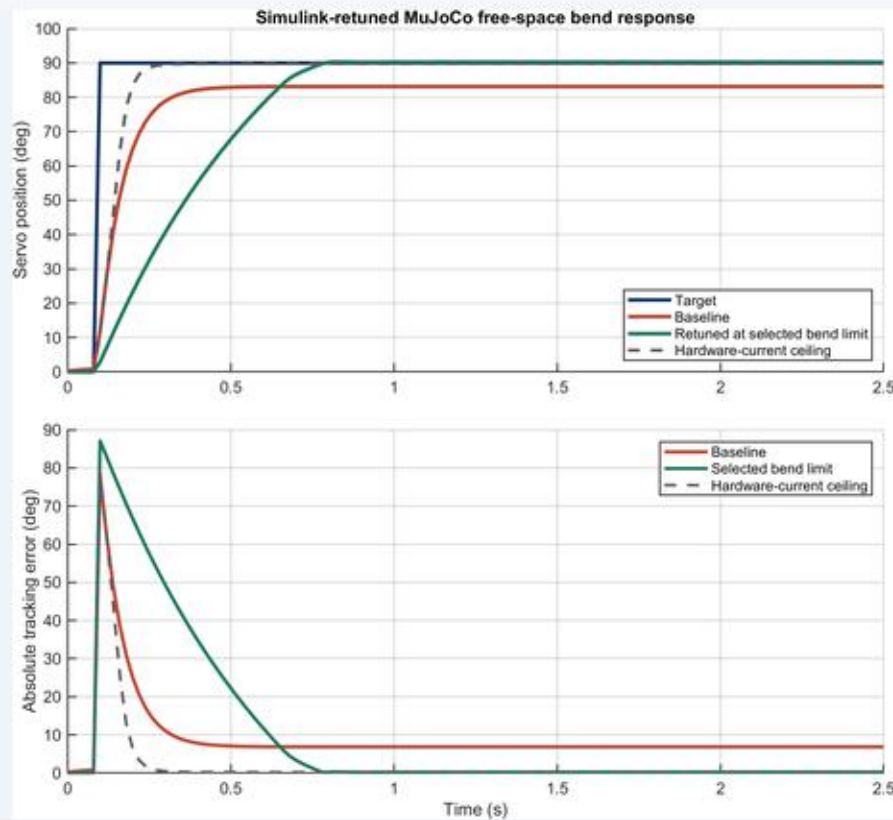
- PhD candidate finishing Summer 2026
- 8+ years robotics and mechatronics R&D
- CAD, controls, firmware, fabrication, and validation



COMPLIANT ROBOTIC HAND

FDM-compatible hand with simulation-to-hardware controls

- Three-motor compliant mechanism
- Simulink/MuJoCo control workflow
- Python Feetech servo tooling and bench demos



Tuning summary

Baseline

Kp 0.90, Ki 0.000, Kd 0.035
Torque limit 0.69 Nm
Final error 6.9 deg
Peak effort 690

Retuned free-space bend

Kp 10.00, Ki 0.800, Kd 0.250
Torque limit 0.16 Nm
Final error 0.3 deg
Peak effort 160

Hardware-current ceiling

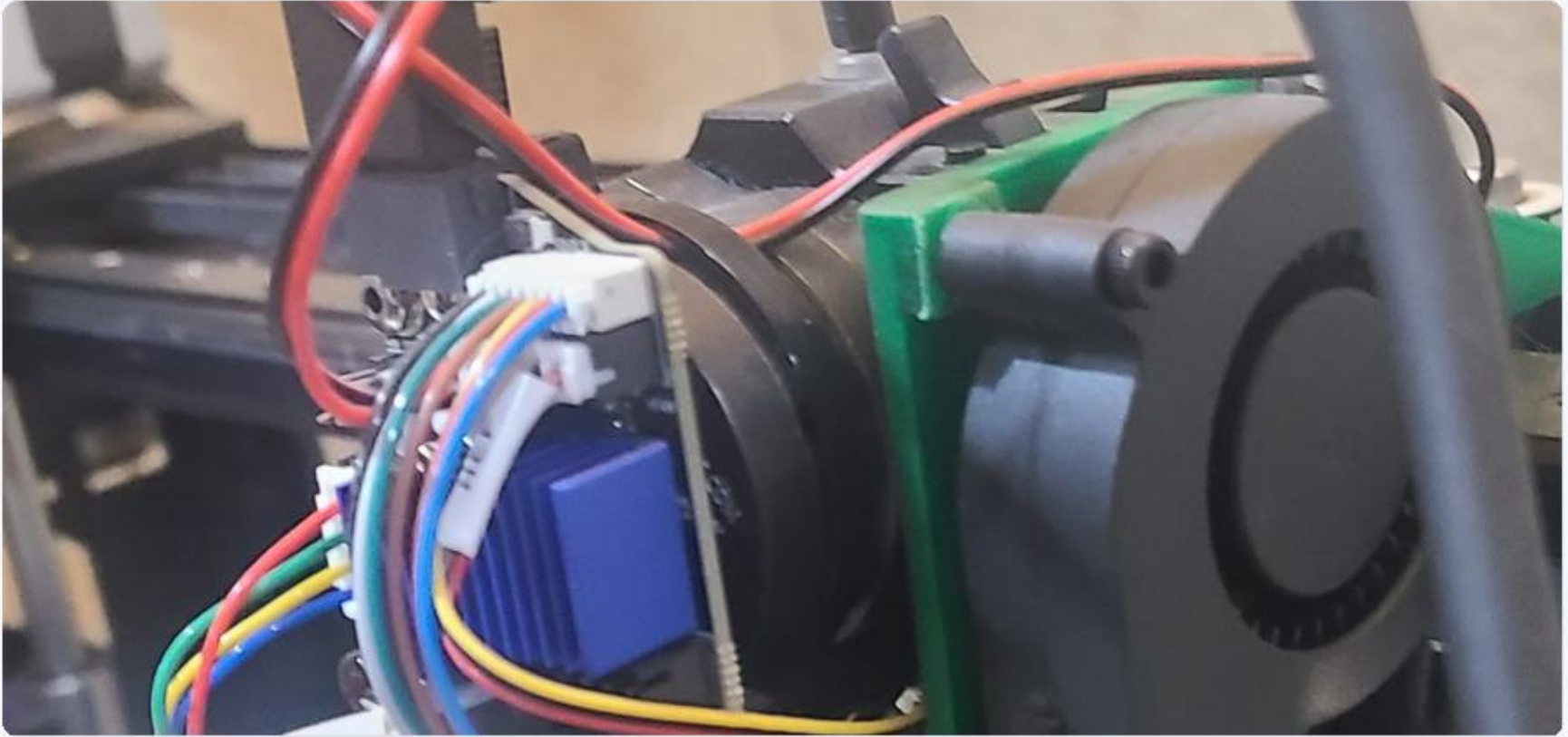
Torque limit 0.69 Nm
Final error 0.2 deg
Peak effort 690

Error change: -6.6 deg
Selected bend limit is not the
final grasp/contact limit.

CONTROLS AND MODELING

Simulink, MuJoCo, and motion-control evidence

- Plant/controller separation
- PID tuning and torque/current limit sweeps
- Robot-learning and dynamics coursework artifacts



PRINTER RETROFIT

Practical machine integration for additive manufacturing

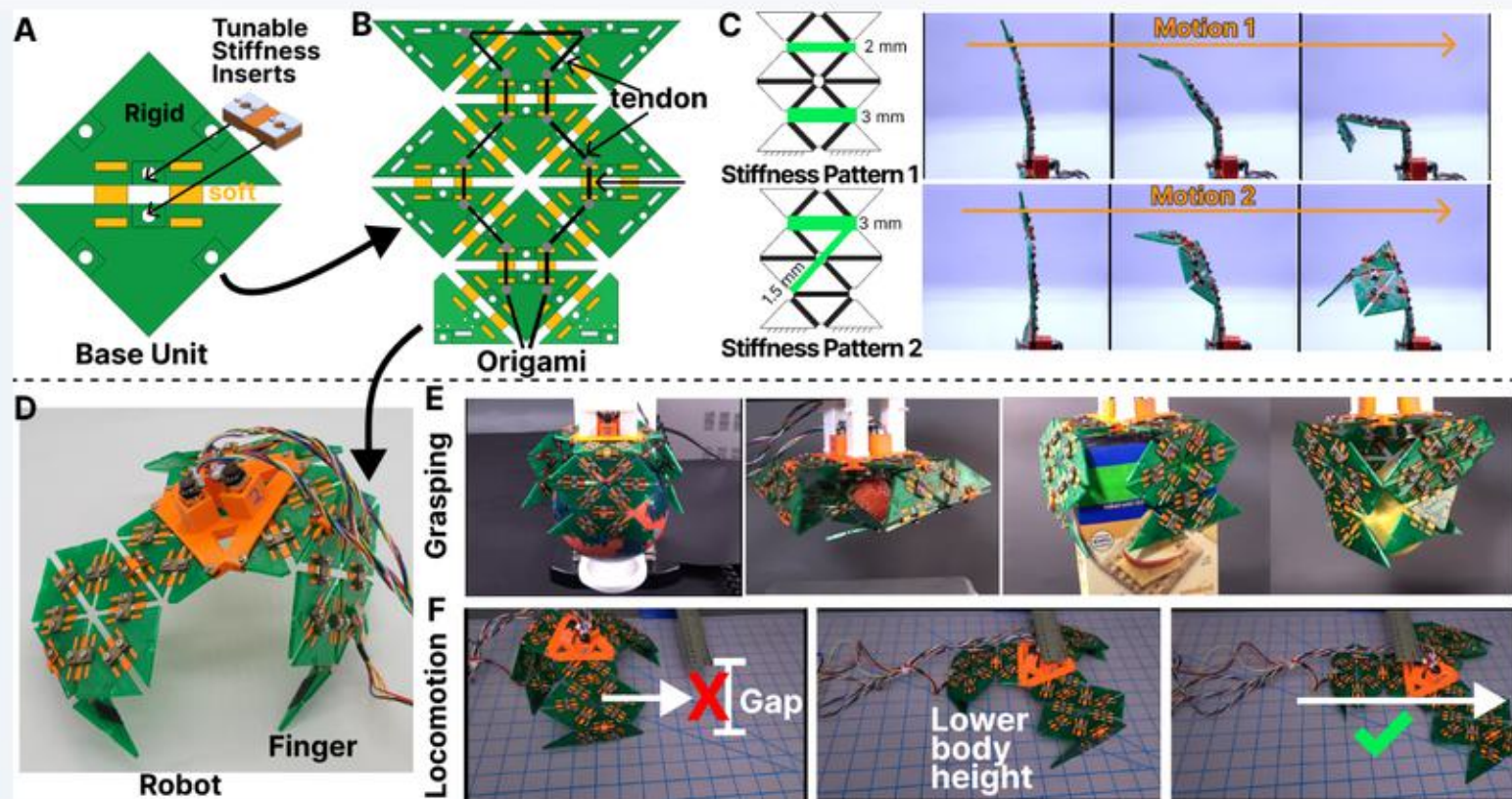
- Right-extruder Smart Orbiter-style retrofit
- CAD packaging and cooling duct design
- Custom Klipper startup G-code



SHAPE-MORPHING ROBOT

Amphibious system integration and validation

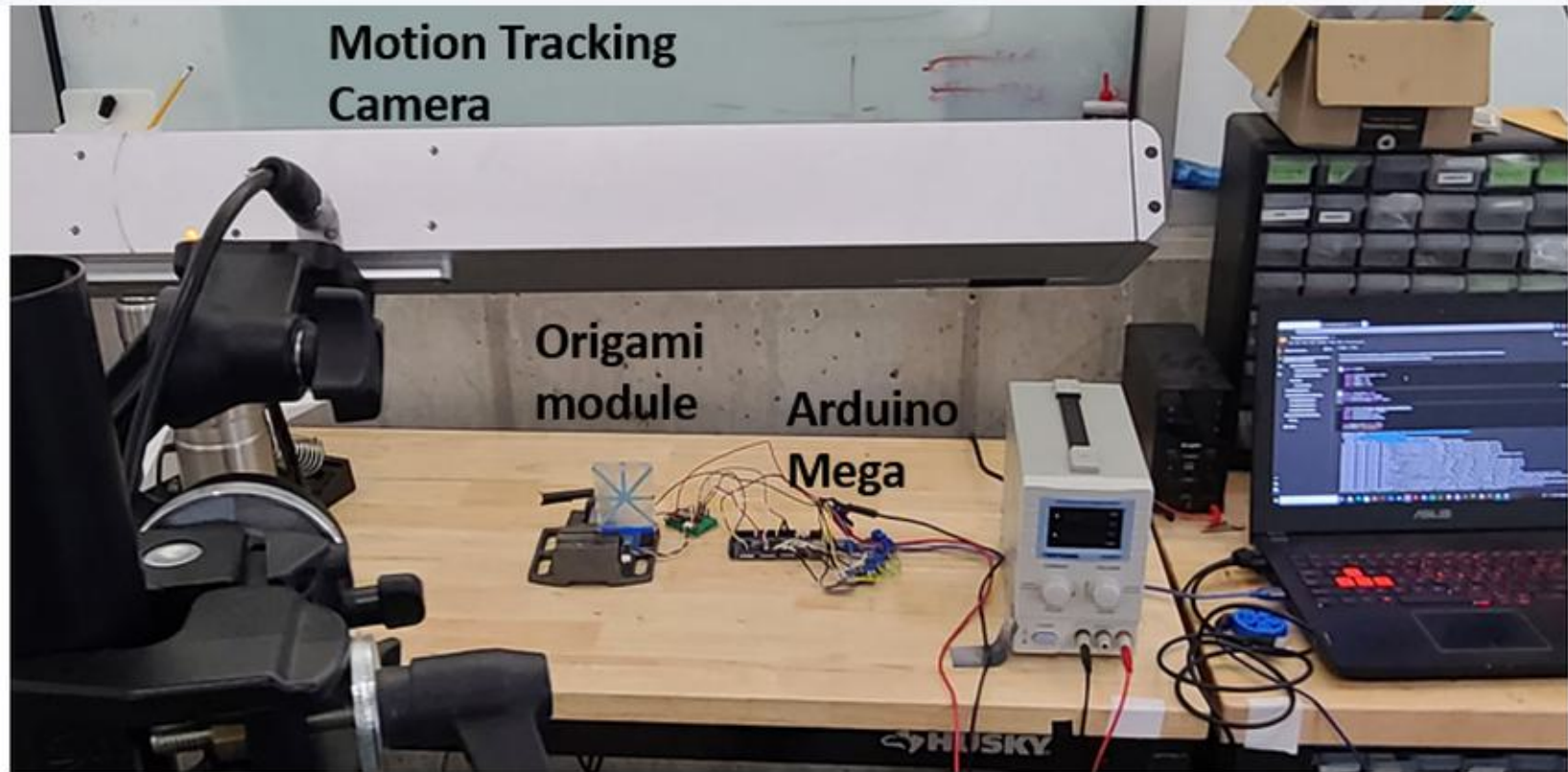
- Waterproof body and morphing limbs
- Encoded motors, BLE modes, thermal actuation
- Tank tests and outdoor land-water transitions



VARIABLE-STIFFNESS ROBOTS

Origami platforms with active stiffness control

- SMP/PETG composites and active VSIs
- Thermistors, heaters, relay isolation, encoders
- Locomotion, gait switching, and adaptive grasping



EXPLORE THE WORK

Portfolio, source evidence, and publications

- Full project pages: elishalerner.github.io
- Source files and control artifacts included
- Available for robotics R&D and mechatronics roles after PhD completion

elishalerner.github.io